

Sakhya Sharma



Adm.No.-25MC0083 +91-7985430070 25mc0083@iitism.ac.in linkedin.com/in/sakhya sharma-12289737a

EXPERIENCE

DST-INSPIRE Summer Research Project(IIT BHU)

Identification and Extraction of Fling-Step Pulses in Earthquake Ground Motions JUNE 2026
– JULY 2026

- Conducted research on the identification and extraction of **fling-step pulses** from near-fault earthquake ground-motion records.
- Analyzed **101 Record Sequence Numbers (RSNs)** from the **PEER Ground Motion Database**.
- Processed **H1, H2, and Vertical components** and analyzed acceleration, velocity, and displacement time histories using **Python**.
- Investigated **long-period velocity pulses and permanent ground displacement** to identify fling-step characteristics.

Research and Web Development Intern(IIT BHU)

Earthquake Data Analysis and Web Application Development [JUNE 2026 – JULY 2026]

- Developed an interactive **web-based earthquake ground-motion analysis application** using data from the **PEER Ground Motion Database**.
- Implemented automated processing of earthquake records to generate the corresponding **time vector, Peak Ground Acceleration (PGA), Peak Ground Velocity (PGV), and Peak Ground Displacement (PGD)**.
- Integrated earthquake data processing and visualization into a web interface, enabling users to analyze ground-motion characteristics interactively.
- Applied **Python, HTML, CSS, and JavaScript** for data processing, visualization, and web application development.
- Designed the application as a practical tool for exploring and interpreting **earthquake ground-motion parameters**.

Research and Web Development Intern

Research and Web Development [JUNE 2026] – [JULY 2026]

- Contributed to research-oriented projects involving **data analysis, technical research, and problem-solving**.
- Developed and maintained web-based applications using **HTML, CSS, and JavaScript**.
- Designed responsive and user-friendly web interfaces and implemented interactive features.
- Applied programming and analytical skills to support research and web development tasks.

.Inspire Exhibition

SEP 2022

Lucknow, Uttar Pradesh, India

- Participating in INSPIRE Science Programme; gained hands on research exposure interacted with scientist and strengthened analytical and teamwork skills .
- INSPIRE Participants experiments that enhanced scientific reasoning innovation mindset, and presentation skills.
- Engaged in laboratory activities and seminars, fostering interest in research and STEM careers.

.Geological Field Training

2nd, DEC 2025 – 11th DEC, 2025

IIT(ISM)Dhanbad, Jharkhand

- Completed Geological Field Training with hands-on experience in lithological mapping structural measurements.

- Visual identification of rocks types based on grain size, texture, and color.
- Measuring of thickness of soil and rock layers using a measuring scale.

BMES WORKSHOP (Bridging Mathematics and Earth Sciences) 16th-18th JAN 2026
IIT(ISM) Dhanbad, Jharkhand

- Modeling and Simulation: Understanding Earth processes using mathematical models.
- Hands-on Training Session: Practical exposure to computational and data-driven techniques in Earth Sciences.
- Data Analysis and Inversion: Quantitative approaches to interpreting geophysical data.

Project

Molecular Logic Gates

Python

JULY 2024 – DEC 2024

- Development of understanding that molecules can perform basic logic operations.
- AND, OR, NOT gates and molecule can behave like tiny logic devices similar to logic gates.
- They respond to chemical inputs and produce an ON or OFF output often in the form of light.

Application of Tensor Analysis in Astrophysics (using Ricci Tensor)

Python

JAN 2025 – APRIL 2025

- Enhanced my understanding of the flattening of space - time by applying tensor analysis through the Ricci Tensor.
- Condition under which curved space approaches flat geometry which produced by matter and energy.

EDUCATION

IIT (ISM), Dhanbad

JUL 2025 – MAY 2028

Pursuing M.Sc.Tech. Applied Geophysics (3rd Sem)

CGPA: 7.7/10 (upto 2nd Sem)

Deen Dayal Upadhyaya Gorakhpur University

JUL 2022 - MAY 2025

B.Sc.[Physics]

CGPA: 7.68/10

Sankar I. C. Gorakhpur

APRIL 2021 - APRIL 2022

12th - 82.4%

SKILLS

- **Programming Web:** Matlab, Python(Basic), R Programming(Basic), HTML, CSS, Java Script, React, Backend
- **MS office:** MS Word, MS Power Point, MS Excel

ACHIEVEMENTS

- Selected among the top 1% students by the Government of Uttar Pradesh
- INSPIRE Scholarship recipient — Department of Science and Technology (DST), Government of India

POSITIONS OF RESPONSIBILITY

- Volunteer, National Service Scheme (NSS), D.D.U. Gorakhpur University
- Class Representative, D.D.U. Gorakhpur University
- Secured 7th rank in INSPIRE Science Exhibition

SPECIAL ELECTIVE COURSES UNDERTAKEN

- National Service Scheme (NSS100), D.D.U. Gorakhpur University
- Disaster Management Mechanism in India (DSS200), D.D.U. Gorakhpur University

- Cyber Laws (LAW203), D.D.U. Gorakhpur University
- Finance and Banking (COMM203), D.D.U. Gorakhpur University